Independent Replication Report: Orbital Hall Effect in Spin-3/2 Hole-Doped Semiconductors (Ge)

Cullen, Wang & Culcer, arXiv:2509.20436v3 (2025/2026)

REPLICATE-PROJECT / TEXTURES-100 — conventional-Kubo rebuild

2026-07-19

VERDICT: PARTIAL (Coverage 5/10, Agreement 5/10)
Headline computed (this work): $\sigma^{L_z}_{xy,\mathrm{conv}} \approx 49\ (\hbar/e)\,\Omega^{-1}\mathrm{cm}^{-1}$ (conventional interband Kubo term only; converged, $< 3\%$ drift over $N = 21/31/41$).
Paper headline: $\sigma_{\mathrm{OHE}} \sim 10^3\ (\hbar/e)\,\Omega^{-1}\mathrm{cm}^{-1}$ (TOTAL: conventional + quantum corrections).

The $\sim 20\times$ gap is *not* a disagreement. The paper's Fig. 2 states explicitly that the quantum corrections $\Delta\sigma_1, \Delta\sigma_2$ *dominate* the total and are what push it to $\sim 10^3$. Our conventional-only rebuild is the correct value of the one term we built; the headline requires the unbuilt quantum-correction terms of the paper's Eq. (6).

1 The paper and its central claim

Cullen *et al.* propose bulk p-type semiconductors (focus: Ge; also Si, GaAs, InAs, InSb) as orbitronic platforms. Using the modern theory of orbital magnetisation applied to the Luttinger–Kohn–Bir–Pikus valence-band Hamiltonian, and *incorporating recently-discovered quantum corrections to the orbital current*, they report an orbital Hall conductivity of order $10^3\ (\hbar/e)\,\Omega^{-1}\mathrm{cm}^{-1}$ — exceeding the spin Hall effect by 2–3 orders of magnitude. The one comparison anchor for this replication is that $\sim 10^3$ headline for Ge, and its decomposition into a conventional piece plus dominant quantum corrections (paper Fig. 2).

Numbered claims checked.

- C1. The 4×4 spherical Luttinger model for Ge has *finite* Berry curvature and OAM even under inversion symmetry, of the monopole-like form $\Omega^m_{i,\mathbf{k}} = -(k_i/k^3)(J_z)_{mm}$ (Eq. 3).
- C2. The orbital Hall conductivity is of the right order for orbitronics; the *total* is $\sim 10^3\ (\hbar/e)\,\Omega^{-1}\mathrm{cm}^{-1}$ (abstract, Fig. 2).
- C3. The quantum corrections $\Delta\sigma_1, \Delta\sigma_2$ are the *dominant* contributions; the conventional interband piece σ_{conv} is sub-dominant (Fig. 2, Sec. text).

2 Governing equations reimplemented

The full model is the 6×6 LKBP Hamiltonian, Eq. (1). For Ge (large split-off gap δ_{so}) the upper 4×4 block suffices, and we use the **spherical approximation**, Eq. (2):

$$H_0 = -\frac{\hbar^2}{2m_0}\Big[(\gamma_1 + \tfrac{5}{2}\bar{\gamma})\, k^2 - 2\bar{\gamma}\,(\mathbf{k}\cdot\mathbf{J})^2\Big], \qquad \bar{\gamma} = \tfrac{1}{2}(\gamma_2+\gamma_3), \tag{1}$$

with $\mathbf{J}$ the spin-3/2 matrices (heavy holes $J_z = \pm 3/2$, light holes $J_z = \pm 1/2$). For Ge we use $\gamma_1 = 13.38$, $\gamma_2 = 4.25$, $\gamma_3 = 5.69 \Rightarrow \bar{\gamma} = 4.97$.

The Berry curvature (Eq. 3) and equilibrium OAM (Eq. 4) are analytic:

$$\Omega^m_{i,\mathbf{k}} = -\frac{k_i}{k^3}(J_z)_{mm}, \qquad L^m_{i,\mathbf{k}} = 3\hbar\bar{\gamma}\,\frac{k_i}{\cdots}(J_z)_{mm}. \tag{2}$$

The orbital current with all matrix elements is Eq. (6), $\langle J^{\alpha}_{\delta}\rangle = \mathrm{Tr}\big[\frac{1}{2}\{\hat{L}^{\alpha},\hat{v}_{\delta}\}\,\rho\big]$. The *conventional* piece keeps only off-diagonal velocity and band-diagonal OAM matrix elements. We implement exactly this conventional interband (intrinsic Kubo / Berry-curvature) term, with the orbital current operator $j^{L_z}_x = \frac{1}{2}\{L_z, v_x\}$:

$$\sigma^{L_z}_{xy,\mathrm{conv}} = \frac{e^2}{\hbar}\,\frac{1}{(2\pi)^3}\int d^3k \sum_{n\,\mathrm{occ}}\sum_{m\neq n} \frac{2\hbar^2\,\mathrm{Im}\big[(j^{L_z}_x)_{nm}(v_y)_{mn}\big]}{(\varepsilon_n-\varepsilon_m)^2}. \tag{3}$$

The velocity operator is $v_i = \frac{1}{\hbar}\partial H_0/\partial k_i$, obtained analytically from Eq. (2). We do **not** build the quantum corrections $\Delta j_{1,2,3}$ of Eq. (6) — this is the deliberate scope of the replication and the source of the coverage cap.

3 Method

Direct numerical Kubo evaluation on a uniform 3D k-grid, spherical 4×4 model:

- Build $H_0(\mathbf{k})$ and its $\mathbf{k}$-gradients analytically; `eigh` at each grid point.
- Rotate v_x, v_y and L_z into the eigenbasis; form $j^{L_z}_x = \frac{1}{2}\{L_z, v_x\}$.
- Sum the interband Berry-curvature-weighted term over occupied hole states ($\varepsilon_n \leq E_F$), degenerate pairs ($|\varepsilon_n - \varepsilon_m| < 10^{-24}$ J) masked.
- Normalise $\int d^3k/(2\pi)^3$; convert SI $\to$ S/cm $\equiv (\hbar/e)\,\Omega^{-1}\mathrm{cm}^{-1}$.

Grid: cubic box $k_{\max} = 4\times10^8\,\mathrm{m}^{-1}$, N^3 points ($N = 21, 31, 41$), $E_F = 10$ meV (also 5 meV). Fermi-surface coverage checked: the heavy-hole Fermi wavevector $k^{hh}_F = \sqrt{2m_0E_F/[\hbar^2(\gamma_1-2\bar{\gamma})]} = 2.76\times10^8\,\mathrm{m}^{-1} < k_{\max}$, so the grid encloses the full occupied hole pocket.

4 Results

5 Comparison and verdict

Our converged conventional-only conductivity, $\sigma^{L_z}_{xy,\mathrm{conv}} \approx 49\,(\hbar/e)\Omega^{-1}\mathrm{cm}^{-1}$, sits $\sim 20\times$ below the paper's $\sim 10^3$ headline. This is *consistent*, not contradictory: the paper's central methodological

Run	N	E_F (meV)	$\sigma^{L_z}_{xy,\mathrm{conv}}$ $(\hbar/e)\Omega^{-1}\mathrm{cm}^{-1}$	t (s)
coarse	21	10	48.30	0.2
med	31	10	49.92	0.9
fine	41	10	49.41	2.4
fine	41	5	33.77	3.8

Table 1: Grid-convergence of the conventional interband OHE for Ge. The $N = 21/31/41$ values (48.3 / 49.9 / 49.4) drift $< 3\% \Rightarrow$ **converged**. Live re-run (2026-07-19, `/home/stevens/comfyui-env/bin/python`) reproduced $N = 31$ to the quoted digits (49.9217).

Quantity	This work	Paper	Status
$\sigma^{L_z}_{xy,\mathrm{conv}}$ (Ge)	≈ 49	sub-dominant term in Fig. 2	consistent (right order fo
σ_{OHE} total (Ge)	not built	$\sim 10^3$	gap = unbuilt $\Delta\sigma$
Finite Ω, L under inversion (C1)	yes, $\Omega_i \propto -k_i/k^3\, J_z$	yes (Eq. 3)	**reproduced**
$\Delta\sigma_{1,2}$ dominate (C3)	not built	yes (Fig. 2)	consistent (our σ_{conv} <

Table 2: This-work vs paper. The conventional term reproduces the paper's physics and order for that term; the total headline requires the quantum corrections, which are out of scope.

point (abstract; Fig. 2; Sec. text lines 590–591 "the quantum corrections $\Delta\sigma_{1,2}$ are the dominant contributions") is precisely that the conventional interband term is sub-dominant and the total is carried by the quantum corrections $\Delta j_1, \Delta j_2$ of their Eq. (6). A tight conventional-only rebuild landing 1–2 orders below the total is the expected signature of that claim. The core single-particle physics — spherical Luttinger bands, finite Berry curvature/OAM under inversion symmetry of the form $\Omega_i = -(k_i/k^3)J_z$, and the right order of magnitude for the conventional piece — **reproduces**. **Verdict: PARTIAL.**

6 Gaps, conventions and scope

1. **Quantum corrections not built.** $\Delta\sigma_1$ (interband-polarization / dipole rotation) and $\Delta\sigma_2$ (interband OAM matrix elements) — the paper's *dominant* terms — are not implemented; nor is $\Delta\sigma_3$ (the $[\hat{r}, \hat{v}]$ term, opposite sign). Closing the headline requires the full Eq. (6) covariant-derivative machinery of the modern theory of orbital magnetisation. This is the coverage cap, and it is a deliberate scope choice, not a physics failure.

2. **Spherical 4×4 approximation only.** Full 4×4 (cubic anisotropy) and 6×6 (split-off band) models, used by the paper for numerical accuracy, are not built. Adequate for Ge per the paper (large δ_{so}), but caps quantitative fidelity.

3. **Conventional-Kubo convention.** We adopt the community-standard Go-*et al.* form (charge-Hall Kubo prefactor $e^2/\hbar$, one velocity replaced by $j^{L_z}_x$, L_z dimensionless). The paper's "proper" (Eq. 7-analogue) current with the tilde/check degenerate/non-degenerate distinction is not separately reconstructed.

4. **Grid & Fermi surface (verified, not a gap).** $k_{\max} > k^{hh}_F$ confirmed; convergence $< 3\%$ over $N = 21/31/41$. The number is trustworthy *for the term computed.*

5. **Extraction tooling degraded (not physics).** `marker`/`nougat` absent; extraction/*.{md,mmd} are honest `pdftotext` interims with hand-transcribed equations.

7 Reproducibility

Interpreter: `/home/stevens/comfyui-env/bin/python` (numpy 2.3.5). Driver: `report/evidence/ohe_spherical`
Evidence JSON: `report/evidence/cullen2025_result.json`.

```
cd /home/stevens/textures-100/corpus/textures-orbital-cullen2025/
/home/stevens/comfyui-env/bin/python work/ohe_spherical.py
# -> writes work/cullen2025_result.json with the N21/31/41 convergence
#    table + _verdict block (grid_covers_FS, computed value, PARTIAL).
```

This `.tex` ships as source; no LaTeX engine is required on the packaging host and it compiles off-host with `pdflatex` (packages: amsmath, booktabs, hyperref, tcolorbox).